be part of the result. These are *true negatives*. In the current scenario, recall is calculated as

```
Recall = 6 / (6+3) x 100% = 66.6%
```

Ideally, we want precision and recall to be a perfect match with no omissions (no relevant documents omitted). However, this is almost impossible because these measures always work against each other. They are inversely proportional: the higher the precision (number of best-match documents), the lower the recall (number of documents returned). Figure 10.3 shows the inverse relationship between these two factors.

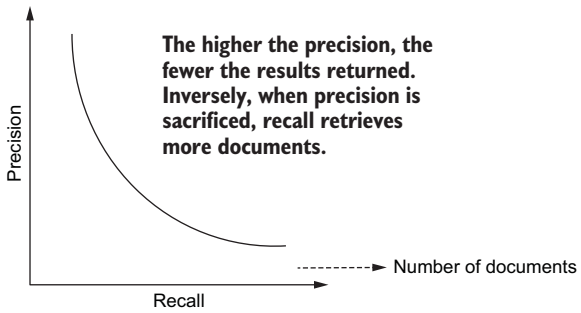


Figure 10.3 Precision and recall are always at odds with each other.

We need to make sure the returned results strike a balance between the precision and recall strategies. Figure 10.4 summarizes precision and recall.

$$\text{Precision} = \frac{\text{Relevant documents retrieved}}{\text{Retrieved documents in the result set}}$$

$$\text{Recall} = \frac{\text{Relevant documents retrieved}}{\text{Relevant documents in the corpus}}$$

Figure 10.4 The formulas for precision and recall

We can tune results for precision and recall when designing and executing queries. We can also use `match` queries, filters, and `boost` to tweak precision and recall to fine-tune their balance. In this chapter we do not directly tune these parameters but rather work on queries to see how the results are affected. If you are interested in learning more about precision and recall in the information retrieval context, I suggest you check out this article from Google for Developers: http://mng.bz/a1eo.